Conditional generation with instantaneous quantum polynomial-time circuits

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Conditional generative models represent families of distributions indexed by an input variable. Adding this dependence to instantaneous quantum polynomial-time (IQP) circuits is constrained by their commuting structure, the input variable must fit within the diagonal block. We introduce state-conditioned IQP Born machines, which instead encode the condition in a classically tractable input state while keeping a single diagonal unitary fixed across the family. The Pauli-Z observables required by the maximum mean discrepancy loss, and gradients, remain efficiently estimable on a classical computer whenever the input state supports efficient sampling and amplitude-ratio evaluation. This retains the classical training framework without sampling from the model during optimization. Experiments on discretized Gaussian families show that one fixed IQP unitary can model multiple conditions and interpolate to values absent from training. Compared with training an independent IQP circuit for each condition, the conditional model requires a single parameter vector (versus one for each condition) and only suffers a small performance penalty. For Curie–Weiss targets, the same framework with an input state that respects the problem symmetry follows the temperature-driven change from a central to a bimodal magnetization profile. In particular, it interpolates at a held-out critical point. We also prove that universal approximation of all continuous conditional laws would require condition states to become asymptotically perfectly distinguishable, which permits operational condition selection. These results establish state conditioning as a route to conditional generative modeling with fixed, classically trainable IQP circuits while separating interpolation from universal approximation.

I. INTRODUCTION

Conditional generative models learn $q_\theta(x \mid c)$, a family of output distributions indexed by a condition c. Quantum circuit Born machines (QCBMs) represent discrete distributions through measurement statistics and can be trained with likelihood-free objectives such as maximum mean discrepancy (MMD) [1–3]. Conditional QCBMs encode c through repeated rotations or dedicated condition qubits [4, 5]. In an instantaneous quantum polynomial-time (IQP) circuit, condition-dependent gates that preserve the IQP form belong to the commuting diagonal block. The diagonal unitary therefore changes with c. We ask whether an IQP Born machine can represent $q_\theta(x \mid c)$ while keeping this unitary fixed and retaining its classical training framework.

Classically trained IQP Born machines separate observable estimation from output sampling. Selected Pauli-Z correlators, character-kernel MMD losses, and their gradients can be estimated classically for a tractable reference state, although sampling from related IQP families is associated with hardness results under standard complexity assumptions [6–9]. We encode c in a classically tractable input state $|\phi(c)\rangle$ and apply one fixed IQP unitary. Product inputs retain the factorized reference distribution used by the Pauli estimator; non-product inputs remain admissible when the state entering the diagonal block supports efficient basis sampling and amplitude-ratio evaluation [6]. The same classical estimators then apply, while the diagonal generators and learned phases remain independent of c. Figure 1 shows this separation. Input-state encodings and their finite Fourier responses are established in quantum circuit learning [10–12]. Here the input state acts as a conditioning interface for a fixed IQP unitary. We use *classically trainable* only for the observable, loss, and gradient estimators; it does not imply classical access to the output distribution.

Gaussian Hamming-weight targets test conditional translation and within-range interpolation. A single IQP unitary models several conditions and responds at values absent from training. It uses one learned phase vector rather than one per condition, with only a small loss in accuracy. Curie–Weiss targets test a change in distribution shape with inverse temperature. With a spin-flip-symmetric input state, the fixed unitary follows the change from a central to a bimodal magnetization profile and improves over an unconditioned control at training and held-out temperatures. It also interpolates at the held-out critical value $\beta = 1$. For the eight-spin system considered here, this is a finite-size crossover rather than an estimate of critical scaling.

State conditioning therefore extends classically trained IQP Born machines from a single distribution to a

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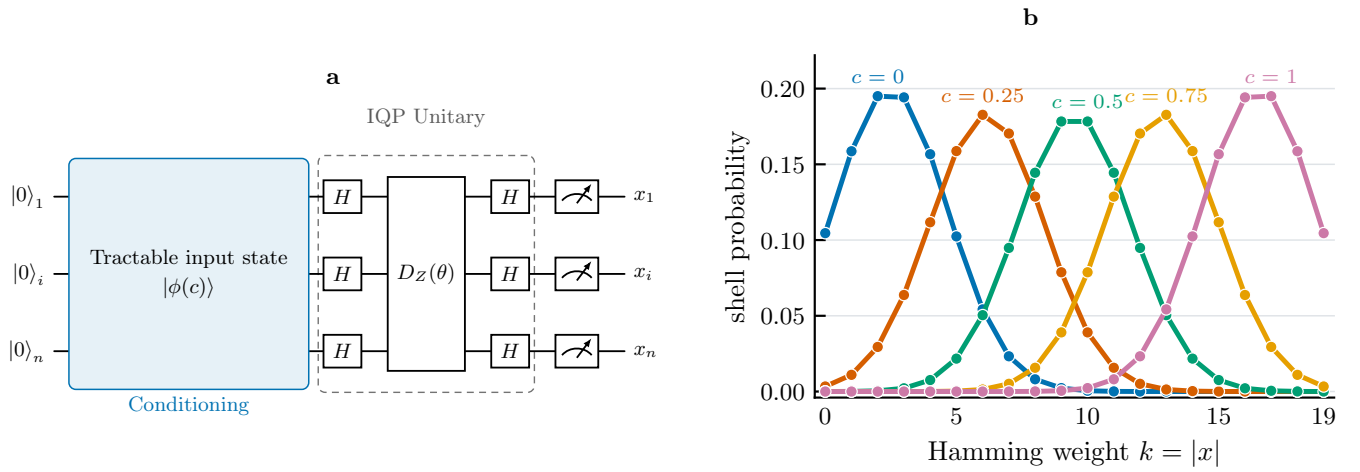


FIG. 1: State conditioning and the Gaussian target family. (a) Conditioning prepares a classically tractable input state $|\phi(c)\rangle$ (a product state in the Gaussian experiment and a spin-flip two-branch state in the Curie–Weiss experiment); the IQP unitary and phase vector are shared across conditions. (b) Five representative 19-qubit target profiles; probability is uniform within each Hamming shell. **[TODO: This is the main figure, we should either make it all about the input state conditioning or also include the Curie-Weiss dataset]**

condition-indexed family without reprogramming the diagonal unitary. Our evidence is limited to within-range interpolation on exchangeable targets. The Curie–Weiss experiment does not isolate the effects of the spin-flip construction and ancilla marginalization. Section III proves that an overlap-preserving, non-multiplexed family cannot approximate every continuous conditional law. We make no claim of classical output sampling, quantum advantage, thermal-state preparation, or favorable optimization behavior [13].

II. CONDITIONAL IQP MODEL

The conditional IQP model has two parts: a condition-dependent input state and a condition-independent unitary. Let $G = (V, E)$ be an undirected graph with $n = |V|$ vertices. Each vertex represents one qubit and one R_Z gate, while each edge represents an R_{ZZ} gate. For $\boldsymbol{\theta} \in \mathbb{R}^{|V|+|E|}$, the graph defines the IQP unitary

$$U_{\boldsymbol{\theta}} = H^{\otimes n} D_G(\boldsymbol{\theta}) H^{\otimes n}, \tag{1}$$

where the diagonal layer is

$$D_G(\boldsymbol{\theta}) = \left[\prod_{\{i,j\}\in E} R_{ZZ}^{(i,j)}(\theta_{ij})\right]\left[\prod_{i\in V} R_Z^{(i)}(\theta_i)\right]. \tag{2}$$

All gates in D_G commute. The unitary has $|V| + |E|$ trainable parameters, one for each vertex and edge.

For each condition c, an encoder prepares a state $|\phi(c)\rangle$ before the IQP unitary. Computational-basis measurement defines

$$q_{\boldsymbol{\theta}}(x \mid c) = |\langle x|U_{\boldsymbol{\theta}}|\phi(c)\rangle|^2, \qquad x \in \{0,1\}^n. \tag{3}$$

Equation (3) highlights the sharing constraint: G and $\boldsymbol{\theta}$ are the same for every condition. The state-preparation circuit need not have IQP form, so the full conditional pipeline is not, in general, a strict IQP circuit. Its learned and compiled diagonal $D_G(\boldsymbol{\theta})$ remains fixed across conditions.

The state entering the diagonal layer is $|\psi(c)\rangle = H^{\otimes n}|\phi(c)\rangle$. Classical estimation of the Pauli-Z expectation values requires samples from $|\langle z|\psi(c)\rangle|^2$ together with ratios between computational-basis amplitudes. This restricts the allowed encoders. Van den Nest's definition of a computationally tractable state, which requires efficient basis sampling and efficient amplitude evaluation, is sufficient [6]. Section IV states the weaker sampling and amplitude-ratio interface used by our estimator.

A. Product-state conditioning

The simplest way to encode the condition is a product state of single-axis rotations,

$$|\phi(c)\rangle = \bigotimes_{i=1}^{n} R_j(\alpha_i(c))\,|0\rangle\,. \tag{4}$$

Here $j \in \{X, Y\}$ denotes the rotation axis and $c \in [0, 1]$ is the normalized condition. Its physical meaning depends on the problem. The condition changes the angles $\alpha_i(c)$ but not the IQP parameters.

[TODO: Maybe this fits better in the experimental setup section] In our experiments we employ R_Y rotations in the encoding and use the following transformation for the input,

$$\alpha_i(c) = \pi(0.35 - 0.70c), \quad i = 1, \ldots, n \tag{5}$$

This affine transformation is chosen heuristically to map the input interval $c \in [0,1]$ symmetrically across the angular range $[0.35\pi, -0.35\pi]$, providing a balanced phase shift that centers the data while avoiding the periodicity bounds of the rotation gate **[TODO: justify better, maybe point towards the results or do some experiment to show that this works better than simply normalizing by pi]**.

Since the angle depends affinely on c, each output probability is a finite Fourier series whose frequencies are signed sums of the encoding frequencies [11, 12].

B. Purified product-state conditioning

Product-state conditioning supplies one pure visible input for each condition. To represent a condition-dependent mixed input, we extend the system with r ancillas and measure only the n visible qubits.

[TODO: maybe this can be reduced to an "ancilla" section as the other encoding also uses ancillas.]

Let $u \in \{0,1\}^r$ label an ancilla state and let $w_c(u)$ be a normalized distribution. Each value of u selects a visible product state

$$|\phi(c,u)\rangle = \bigotimes_{i=1}^{n} R_{j_i}(\alpha_i(c,u))\,|0\rangle\,. \tag{6}$$

The joint input

$$|\Phi(c)\rangle = \sum_{u\in\{0,1\}^r} \sqrt{w_c(u)}\,|u\rangle_a\,|\phi(c,u)\rangle_v \tag{7}$$

is a purification of the visible mixture $\sum_u w_c(u)\,|\phi(c,u)\rangle\langle\phi(c,u)|$. Let $N = n + r$ and let $U_{\boldsymbol{\theta}}^{(N)}$ be an IQP unitary on the joint register. The visible output distribution is

$$q_{\boldsymbol{\theta}}(x \mid c) = \sum_{z\in\{0,1\}^r} \left|\langle x, z|U_{\boldsymbol{\theta}}^{(N)}|\Phi(c)\rangle\right|^2. \tag{8}$$

The ancillas can therefore change the rank of the visible state. If the graph contains visible–ancilla edges, they also change the interaction structure of the IQP unitary. These are different sources of model capacity and should be distinguished in comparisons. Classical training remains possible only when the joint state entering the diagonal layer supports the sampling and amplitude-ratio operations defined in Section IV.

C. Symmetry-informed conditioning

[TODO: Simplify so its easier to understand]

Conditional models can exploit symmetries that relate different conditions, as well as symmetries within one target distribution. Let a finite group Γ act on both conditions and bit strings. We want the model to satisfy

$$q_\theta(g\cdot x \mid g\cdot c) = q_\theta(x \mid c), \qquad g \in \Gamma. \tag{9}$$

If the group does not change c, Eq. (9) reduces to the usual invariance $q_\theta(g\cdot x \mid c) = q_\theta(x \mid c)$. If instead $c = h$ is a signed field and g flips every bit, the same equation becomes $q_\theta(\bar{x} \mid -h) = q_\theta(x \mid h)$.

Let S_g implement the action on bit strings, $S_g\,|x\rangle = |g\cdot x\rangle$. Equation (9) follows from two requirements:

$$[U_\theta, S_g] = 0, \qquad |\phi(g\cdot c)\rangle = e^{i\omega_g(c)} S_g\,|\phi(c)\rangle\,. \tag{10}$$

The first requirement makes the shared IQP unitary equivariant. The second requires the encoder to transform conditions and output states consistently. The phase $e^{i\omega_g(c)}$ disappears upon measurement, and hence

$$\begin{aligned} q_\theta(g\cdot x \mid g\cdot c) &= \left|\langle x|S_g^\dagger U_\theta|\phi(g\cdot c)\rangle\right|^2 \\ &= \left|\langle x|U_\theta|\phi(c)\rangle\right|^2. \end{aligned} \tag{11}$$

One way to construct such an encoder is to start from a base state $|\eta(c)\rangle$ and average it over the joint group action,

$$|\phi_\Gamma(c)\rangle = \frac{1}{\sqrt{\mathcal{N}(c)}} \sum_{g\in\Gamma} S_g\,|\eta(g^{-1}\cdot c)\rangle\,, \tag{12}$$

where $\mathcal{N}(c)$ normalizes the state. The group fixes how conditions and outputs transform, while $|\eta(c)\rangle$ controls the remaining dependence on c.

For a permutation π of the qubits, the commutation requirement holds when relabeling every qubit i as $\pi(i)$ leaves the diagonal layer unchanged. Concretely, $\{i,j\} \in E$ if and only if $\{\pi(i),\pi(j)\} \in E$, and gates mapped into one another use the same angles: $\theta_i = \theta_{\pi(i)}$ and $\theta_{ij} = \theta_{\pi(i)\pi(j)}$. Every permutation preserves the edge set of a complete graph. The corresponding unitary is invariant under all qubit permutations when all vertex angles are equal and all edge angles are equal.

Symmetry does not imply classical tractability. Even if the base state $|\eta(c)\rangle$ is easy to prepare, the coherent sum in Eq. (12) may be hard to sample or query. A useful case occurs when the amplitudes depend only on a polynomial number of classes, such as the $n+1$ Hamming-weight values. One can then sample the class first and evaluate amplitude ratios from the classes of z and $z \oplus a$.

III. LIMITS OF CONDITIONAL UNIVERSALITY

Classical tractability controls access to the estimator, not conditional expressivity. Let ρ_c be the input density operator on every visible, hidden, and ancillary qubit. For fixed parameters θ, the IQP unitary, partial trace, and visible measurement define a quantum-to-classical channel $\mathcal{M}_\theta$, so $q_\theta(\cdot \mid c) = \mathcal{M}_\theta(\rho_c)$. Write $D_{\mathrm{tr}}(\rho,\sigma) = \frac{1}{2}\|\rho - \sigma\|_1$.

Proposition 1 (Pairwise distinguishability bound)**.** *For any target conditional law p and any two conditions c, c',*

$$\sup_t \mathrm{TVD}(q_\theta(\cdot \mid t), p(\cdot \mid t)) \geq \frac{1}{2}\left[\mathrm{TVD}(p(\cdot \mid c), p(\cdot \mid c')) - D_{\mathrm{tr}}(\rho_c, \rho_{c'})\right]_+, \quad (13)$$

where $[u]_+ = \max\{u, 0\}$.

Measurement cannot increase trace distance. Combining this contraction with the triangle inequality proves Proposition 1; Appendix A gives the full derivation. An exactly universal conditional family must represent target laws with disjoint support at any chosen pair of conditions. Equation (13) then requires $D_{\mathrm{tr}}(\rho_c, \rho_{c'}) = 1$ for every $c \neq c'$, which is equivalent to orthogonal input supports.

The same obstruction survives resource scaling. We call an encoder family overlap preserving, and hence non-multiplexed, if some $c \neq c'$ and $\gamma > 0$ satisfy

$$D_{\mathrm{tr}}\left(\rho_c^{(R)}, \rho_{c'}^{(R)}\right) \leq 1 - \gamma \quad \text{for every allowed resource } R. \quad (14)$$

A total-variation-continuous target can interpolate from δ_{0^n} at c to δ_{1^n} at c'. Every model in the family then has uniform error at least $\gamma/2$. Thus universal approximation of all continuous conditional laws forces every distinct pair of condition states to become asymptotically perfectly distinguishable. Such states are operational labels in the limit: their discrimination error tends to zero, enabling asymptotically reliable condition routing.

Coherent selector notation does not remove this routing structure. If a factorized selector state $\sigma_S(c)$ controls a joint IQP diagonal $D = \sum_s |s\rangle\langle s| \otimes D_s$ and S is discarded, then

$$q(x \mid c) = \sum_s \langle s|\sigma_S(c)|s\rangle q_s(x). \quad (15)$$

The off-diagonal selector coherences vanish under the partial trace. Selector and Bernstein constructions therefore produce a classical mixture of condition-independent IQP branches rather than a shared universal conditional mechanism.

For the broadcast product encoder with N copies of $R_Y(\alpha(c))\,|0\rangle$, the input trace distance is

$$d_N(c, c') = \sqrt{1 - \cos^{2N}\left(\frac{\alpha(c) - \alpha(c')}{2}\right)}. \quad (16)$$

If α is L_α-Lipschitz, a continuous target that changes between two deterministic strings over an interval of width h has uniform error at least

$$\frac{1}{2}\left(1 - \frac{\sqrt{N} L_\alpha h}{2}\right)_+. \quad (17)$$

At the Gaussian setting $N = 19$, $L_\alpha = 0.70\pi$, and $h = 0.01$, this certificate equals 0.4760. It is a worst-case expressivity result, not a bound on the reported Gaussian targets. Fixed Fourier bandwidth and parameter dimension impose separate restrictions. In particular, the complete two-local models have 190 phases versus $5(2^{19} - 1) = 2{,}621{,}435$ conditional-table dimensions in the five-condition Gaussian setting, and 91 phases versus $6(2^8 - 1) = 1{,}530$ dimensions at the six Curie–Weiss training temperatures. Appendix A proves each claim and states the narrower compatibility condition left open by this no-go result.

IV. CONDITIONAL IQP TRAINING

For training we employ the maximum mean discrepancy (MMD) loss [14] adapted to the conditional generation setting. The MMD is an integral probability metric defined by a positive-definite kernel k. For two distributions p and q on $\{0,1\}^n$, the squared MMD is

$$\mathrm{MMD}^2(p, q) = \mathbb{E}_{x,x' \sim p}[k(x, x')] - 2\mathbb{E}_{x \sim p,\, y \sim q}[k(x, y)] + \mathbb{E}_{y,y' \sim q}[k(y, y')]. \quad (18)$$

We use the Gaussian kernel, which for binary strings is

$$k_\sigma(x, y) = \exp[-d_H(x, y)/(2\sigma^2)], \quad (19)$$

where d_H is the Hamming distance. Let $a \in \{0,1\}^n$ have independent coordinates

$$a_i \sim \mathrm{Bernoulli}(q_\sigma), \qquad q_\sigma = \frac{1 - e^{-1/(2\sigma^2)}}{2}. \quad (20)$$

Writing $\chi_a(x) = (-1)^{a \cdot x}$, the kernel has the character representation

$$k_\sigma(x, y) = \mathbb{E}_a[\chi_a(x)\chi_a(y)]. \quad (21)$$

Define the character moment of a distribution p by $\mu_p(a) = \mathbb{E}_{x \sim p}[\chi_a(x)]$. Substituting Eq. (21) into Eq. (18) gives

$$\mathrm{MMD}^2_\sigma(p, q) = \mathbb{E}_a\left[(\mu_p(a) - \mu_q(a))^2\right]. \quad (22)$$

For a Born machine, $\mu_q(a)$ is the expectation value of the Pauli string $Z_a = \bigotimes_i Z_i^{a_i}$. The MMD can therefore be estimated by sampling Pauli-Z moments rather than bit strings from the model.

The training objective balances the per-condition losses equally over the training conditions C,

$$\mathcal{L}(\theta) = \frac{1}{|C|}\sum_{c \in C}\frac{1}{|\Sigma|}\sum_{\sigma \in \Sigma}\mathrm{MMD}^2_\sigma\big(p(\cdot \mid c),\, q_\theta(\cdot \mid c)\big). \quad (23)$$

We next state the classical estimator for these moments. Let

$$\psi_c(z) = \langle z|H^{\otimes n}|\phi(c)\rangle \quad (24)$$

be the computational-basis amplitudes of the state entering D_G. The encoder is admissible when, uniformly over c, a classical algorithm can sample z from $|\psi_c(z)|^2$ and evaluate

$$R_c(z,a) = \frac{\psi_c(z \oplus a)}{\psi_c(z)} \tag{25}$$

for sampled z and every mask a used by the loss. The ratio need only be defined on the support of $|\psi_c|^2$. A sampler without amplitude-ratio access is not sufficient.

Let $f_{\boldsymbol{\theta}}(z)$ be the phase defined by $D_G(\boldsymbol{\theta})\,|z\rangle = e^{i f_{\boldsymbol{\theta}}(z)}\,|z\rangle$ and set

$$\Delta f_{\boldsymbol{\theta}}(z,a) = f_{\boldsymbol{\theta}}(z \oplus a) - f_{\boldsymbol{\theta}}(z). \tag{26}$$

Conjugating Z_a through the final Hadamards gives X_a, and hence

$$\begin{aligned} \mu_{q_{\boldsymbol{\theta}}}(a \mid c) &= \langle \psi(c) | D_G^\dagger X_a D_G | \psi(c) \rangle \\ &= \mathrm{Re}\, \mathbb{E}_{z \sim |\psi_c|^2} \left[R_c(z,a) e^{i \Delta f_{\boldsymbol{\theta}}(z,a)} \right]. \end{aligned} \tag{27}$$

The phase difference is computed directly from the gates in D_G, with cost linear in $|V| + |E|$. Product-state encoders provide the required samples and ratios by factorizing over qubits. Other encoders are allowed when they expose the same two operations.

The square in Eq. (22) must be estimated without squaring a single noisy sample mean. For a fixed condition, bandwidth, and mask, let

$$Y_s = \chi_a(x_s) - \mathrm{Re}\left[R_c(z_s,a) e^{i \Delta f_{\boldsymbol{\theta}}(z_s,a)} \right], \tag{28}$$

where $x_s \sim p(\cdot \mid c)$ and $z_s \sim |\psi_c|^2$ are independent. Then $\mathbb{E}[Y_s] = \mu_p(a \mid c) - \mu_{q_{\boldsymbol{\theta}}}(a \mid c)$, and the U-statistic

$$\widehat{\Delta^2}(a,c) = \frac{1}{S(S-1)} \sum_{s \neq t} Y_s Y_t \tag{29}$$

is unbiased for the squared moment discrepancy. Sampling c, σ, and a according to Eqs. (23) and (20) gives an unbiased finite-sample estimate of the conditional loss.

The encoder maps are fixed during this optimization, so $|\psi_c(z)|^2$ and $R_c(z,a)$ do not depend on $\boldsymbol{\theta}$. Gradients with respect to the IQP phases follow by differentiating only $\Delta f_{\boldsymbol{\theta}}$ in the sampled expression. If the encoder contains trainable parameters that change its sampling distribution, this argument no longer applies without an additional gradient estimator.

The amplitude ratio also gives a simple second-moment bound,

$$\mathbb{E}_{z \sim |\psi_c|^2} \left[|R_c(z,a)|^2 \right] = \sum_{z : \psi_c(z) \neq 0} |\psi_c(z \oplus a)|^2 \leq 1. \tag{30}$$

The single-sample Pauli estimator therefore has variance at most one. This is not a zero-variance result and does not guarantee favorable optimization. For an ancilla model, the same derivation applies to the joint state. Visible masks are padded with zeros on the unmeasured ancillas.

Classical trainability in this work refers only to estimating the observables, loss, and phase gradients. It does not provide efficient classical samples from $q_{\boldsymbol{\theta}}(x \mid c)$ or efficient exact evaluation of its probabilities.

V. EXPERIMENTAL SETUP

A. Research questions and confirmation policy

The experiments address four questions. First, can one fixed IQP unitary model a family of distributions when only its input state depends on the condition? Second, does the learned response interpolate to conditions excluded from training and checkpoint selection? Third, which part of an ancilla-assisted construction supplies useful capacity: the rank of the visible input, the visible–ancilla interactions, or the additional trainable phases? Finally, can an equivariant encoder transfer information between conditions related by a known symmetry?

We separated model development from confirmation. Earlier runs were used to select the encoder families, condition maps, optimization schedule, and ancilla counts. We then froze the protocols below and assigned new data and optimizer seeds to every confirmatory fit. We report every declared run, including negative and mixed comparisons. All conditions called held out lie strictly inside the training interval; we therefore use *interpolation* rather than extrapolation throughout.

B. Target families and condition splits

The first target is a discrete Gaussian over the Hamming weight $K(x) = \sum_i x_i$. For condition $c \in [0,1]$, the target mean is $\mu(c) = n(0.15 + 0.70c)$ and the nominal width is $w\sqrt{n}$. We calibrate the location of the truncated Gaussian on $K \in \{0, \ldots, n\}$ so that its mean is exactly $\mu(c)$, then distribute the shell probability uniformly over the $\binom{n}{k}$ strings of weight k. The factorial study uses

$$\begin{aligned} n &\in \{8, 12, 16, 19\}, \qquad w \in \{0.35, 0.50, 0.75\}, \\ C_{\mathrm{train}} &= \{0, 0.25, 0.50, 0.75, 1\}, \\ C_{\mathrm{test}} &= \{(2j+1)/16 : j = 0, \ldots, 7\}. \end{aligned} \tag{31}$$

At $n = 19$ and $w = 0.50$, we also use nested grids containing $K \in \{2, 3, 5, 9\}$ training conditions. One allocation keeps 10,000 training and 2,000 validation strings per condition. The other divides fixed totals of 18,000 and 3,600 strings among the K conditions.

The second target is the finite Curie–Weiss law

$$p_{\beta,h}(x) = \frac{1}{Z(\beta,h)} \exp\left[\frac{\beta}{2n} M(x)^2 + hM(x)\right], \qquad M(x) = \sum_{i=1}^{n} (2x_i - 1). \tag{32}$$

The temperature study sets $h = 0$, uses $n \in \{8, 12, 16\}$, trains at $\beta \in \{0.5, 0.7, 0.9, 1.1, 1.3, 1.5\}$, and evaluates at the interleaved temperatures $\{0.6, 0.8, 1.0, 1.2, 1.4\}$. The ancilla mechanism comparison focuses on the critical region $\beta \in \{0.8, 1.0, 1.2\}$ but reports the full temperature curve. A separate field study fixes $\beta = 1$, trains at $h \in \{0, 0.15, 0.30\}$, and tests at $h = \pm 0.075$ and ± 0.225. Reflected training fields -0.15 and -0.30 test symmetry transfer without adding negative-field samples to the one-sided training set.

The applied benchmark uses the public Muonic Force Carrier event samples at incoming energies $E \in \{50, 75, 100, 125, 150, 175, 200\}$ GeV [4]. We discretize the scaled outgoing-muon energy into 16, 32, and 64 bins and encode the bin index using either ordinary binary or fixed Gray coding. For each representation, 10,000 events per observed energy form the training set, 2,000 form the validation set, and the remaining events are reserved for final evaluation. Five folds omit one interior energy in turn, $E \in \{75, 100, 125, 150, 175\}$ GeV. Before conditional training, three development seeds screen independent fits at 50, 125, and 200 GeV. We continue a coding–resolution cell only if every anchor has median validation TVD at most 0.10 and no seed exceeds 0.15. Failed screens remain part of the reported experiment.

C. Models and controls

Unless stated otherwise, the IQP diagonal contains one R_Z gate per qubit and one R_{ZZ} gate per edge of the complete graph. A model on N qubits therefore has $N(N+1)/2$ trainable phases. The Gaussian and MFC conditioned models use the product-state encoder in Eq. (4) with the affine map in Eq. (5). Their no-condition controls receive the neutral input $c = 1/2$ for every target while retaining the same circuit, data, and optimization budget. Independent single-condition circuits provide a capacity reference. At a held-out Gaussian condition we also compare against the better adjacent stored output and against the target-blind barycentric mixture of the two adjacent outputs. For MFC, the separate circuit is an oracle capacity reference fitted to the omitted energy's training split; it is excluded from held-out-condition generalization claims.

For the zero-field Curie–Weiss family, the $m = 0$ controls use either the product input or the fixed spin-flip two-branch input. The ancilla study adds $r \in \{1, 3, 5\}$ qubits and compares three constructions at every visible size. The *full* model uses the complete graph on all $n + r$ qubits. The *disconnected* model uses the same purified input but removes every visible–ancilla edge, which isolates the effect of the visible input rank. The *matched visible* control removes the ancillas and adds a fixed set of distinct three-qubit diagonal generators until its phase count equals that of the full model. These three-local terms are used only as a capacity control; they are not part of the graph ansatz defined in Section II. No-condition controls are evaluated at $r = 0$ and $r = 5$.

For the field experiment, let

$$|\eta(h)\rangle = \bigotimes_{i=1}^{n} R_Y[-7\pi h/6]\,|0\rangle\,. \tag{33}$$

The symmetry-informed model uses

$$|\phi_{\mathrm{eq}}(h)\rangle = \frac{|\eta(h)\rangle + X^{\otimes n}\,|\eta(-h)\rangle}{\sqrt{\mathcal{N}(h)}}. \tag{34}$$

It obeys $|\phi_{\mathrm{eq}}(-h)\rangle = X^{\otimes n}\,|\phi_{\mathrm{eq}}(h)\rangle$ and hence enforces $q_\theta(\bar{x} \mid -h) = q_\theta(x \mid h)$. We compare it with an unconstrained product encoder trained on the same nonnegative fields, the same product encoder trained on both signs, and a spin-flip-invariant encoder that incorrectly imposes $q(x \mid h) = q(\bar{x} \mid h)$ at nonzero field.

The MFC benchmark adds two classical references. Linear interpolation between the adjacent empirical training histograms gives a target-blind nonparametric baseline. A categorical neural network takes the normalized incoming energy as input, applies two width-32 hidden layers, and predicts the bin probabilities with a softmax output. We treat these as contextual references, not as targets that the IQP model must outperform.

D. Training and evaluation

All IQP fits optimize only the condition-independent phases $\boldsymbol{\theta}$; the encoder maps remain fixed. We use Adam for 2,000 updates with cosine learning-rate decay from 10^{-2} to 10^{-3}. The loss averages the Hamming kernels with $\sigma \in \{1, 2, 4\}$. Each training estimate uses 64 character masks and 8,192 reference samples per bandwidth. Every 25 updates, a fixed validation estimate with 256 masks and 16,384 reference samples selects the checkpoint with the lowest conditional MMD. Held-out conditions and final target probabilities never enter checkpoint selection. The categorical MFC baseline minimizes cross entropy under the same training–validation split and selects its checkpoint by validation loss.

For the synthetic targets, exact enumeration after checkpoint selection gives the bitstring TVD, Hamming-profile TVD, conditional moments, and exact MMD. For an ancilla model we first sum the joint probabilities over the unmeasured qubits. The MFC evaluation decodes the predicted bit strings to energy bins and reports TVD, one-dimensional Wasserstein distance, and errors in the first two scaled-energy moments. The field study also

reports the equivariance defect

$$\epsilon_{\mathrm{eq}}(h) = \frac{1}{2}\sum_{x} |q_\theta(x \mid h) - q_\theta(\bar{x} \mid -h)| \,. \tag{35}$$

The supplementary estimator audit compares moments, losses, and gradients against exact calculations at sample budgets $S \in \{256, 1024, 4096, 16384\}$. A separate topology ablation compares complete, rectangular-grid, and heavy-hex graphs at $n = 19$, $w = 0.50$, and five training conditions.

E. Statistical analysis

The confirmatory campaign uses ten fresh optimizer seeds, numbered 100 through 109, with matched data draws wherever targets are sampled. We report the median and interquartile range across seeds. Method effects are computed within each seed after averaging over the declared conditions, then summarized by a deterministic paired bootstrap with 10,000 resamples and a 95% interval. The seed, rather than an individual condition or bit string, is the replication unit. These are cellwise intervals; the frozen protocol does not apply a familywise correction across contrasts.

For Gaussian interpolation, we require the conditioned model to improve on the no-condition control at held-out conditions and allow at most 0.05 added TVD relative to the interior training conditions. The adjacent lookup and mixture comparisons are reported without a required win. The ancilla analysis tests the three paired differences identified above rather than combining them into one effect. The equivariant field model must improve on one-sided unconstrained training at negative fields and remain within 0.02 TVD of the two-sided model. For MFC data, the generalization claim is restricted to the comparison with the no-condition IQP control. We make no empirical claim about extrapolation, efficient classical output sampling, or quantum advantage.

VI. RESULTS

[TODO: Insert the Gaussian confirmatory results before the Curie–Weiss subsection.]

A. Ancilla-assisted Curie–Weiss interpolation

In the tested ladder, the full ancilla model interpolates across the finite Curie–Weiss crossover once the ancilla register contains at least three qubits. At the three held-out temperatures nearest the transition, the $r = 5$ model attains median profile TVDs of 0.051, 0.058, and 0.069 for $n = 8$, 12, and 16, respectively. The corresponding bitstring TVDs are 0.075, 0.098, and 0.132. Thus the model retains the magnetization profile as the visible system grows, although its error on individual bit strings increases more strongly with size.

Conditioning accounts for most of this accuracy. The $r = 5$ no-condition control has the same generator count and optimization budget but receives the neutral input at every temperature. Its median critical-region profile TVD increases from 0.117 at $n = 8$ to 0.198 at $n = 16$, compared with 0.051 to 0.069 for the conditioned model. Bitstring TVD separates the models more sharply at the two larger sizes: the conditioned and no-condition values are 0.098 versus 0.236 at $n = 12$, and 0.132 versus 0.312 at $n = 16$. The same ordering holds when all five held-out temperatures are included.

The interpolation error is not uniform in temperature. For the $n = 16$, $r = 5$ model, median profile TVD is 0.058, 0.038, and 0.117 at the held-out temperatures $\beta = 0.8$, 1.0, and 1.2. The model therefore represents the central held-out point accurately but overshoots the growth of the ordered modes on the colder side of the crossover. The same pattern is visible in the magnetization and susceptibility: both follow the target response, but their largest critical-region bias occurs at $\beta = 1.2$ and grows with n. These finite systems test interpolation through a crossover; they do not estimate a critical exponent or the thermodynamic critical point.

The paired mechanism comparisons explain why the larger ancilla models work. Relative to the spin-flip $m = 0$ model, the disconnected construction improves bitstring TVD at every size and ancilla count. The corresponding profile-TVD effect is also positive except for the inconclusive $n = 16$, $r = 1$ cell. Once $r \geq 3$, restoring visible–ancilla interactions improves both metrics in all twelve combinations of size, ancilla count, and metric. At $r = 5$, the profile-TVD effects range from 0.024 to 0.027, while the bitstring-TVD effects range from 0.061 to 0.087. The full model also beats the parameter-matched visible control for both metrics at $r = 3$ and $r = 5$. Hence neither additional phases alone nor a mixed visible input without cross-register interactions accounts for the full result.

The $r = 1$ cells give a useful boundary rather than a smaller version of the same effect. Adding visible–ancilla interactions to the disconnected $r = 1$ model worsens profile TVD at all three sizes. Against the parameter-matched visible model, the full $r = 1$ model improves bitstring TVD but worsens profile TVD at $n = 8$ and $n = 12$ and gives an inconclusive profile difference at $n = 16$. In this comparison, the single-ancilla model redistributes probability among individual strings without improving the magnetization-shell probabilities. The confirmatory evidence supports the combined $r = 3$ and $r = 5$ construction, not a monotone claim that every added ancilla or interaction is beneficial.

VII. DISCUSSION

Conditional generation with IQP Born machines represents a promising avenue for quantum generative modeling as it combines the scalability of IQP circuits and the utility of more general models that enable conditional generation...

VIII. CONCLUSIONS

The main contribution of this work is the conditional formulation of classically trained IQP Born machines. The contraction result fixes the scope of this formulation: a non-multiplexed encoder cannot be universal for every continuous conditional law. A positive approximation theorem must instead restrict the target family to distributions whose pairwise total-variation geometry is compatible with the trace-distance geometry of the input encoding.

DATA AVAILABILITY

The code and data supporting the findings of this study are openly available in Zenodo [**?**].

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Appendix A: Conditional universality no-go theorem

This appendix gives the complete derivation of Section III. The result applies to density operators, mixed visible states, hidden qubits, and arbitrary trainable quantum channels. The IQP structure enters only in the selector and fixed-bandwidth specializations.

1. Channel formulation and contraction

Let ρ_c be the state supplied at condition c on all model registers. For fixed circuit parameters θ, the IQP unitary, hidden-register trace, and visible measurement form a quantum-to-classical channel

$$\mathcal{M}_\theta(\rho)(x) = \mathrm{Tr}\left(M_x^{(\theta)}\rho\right), \qquad q_\theta(x \mid c) = \mathcal{M}_\theta(\rho_c)(x), \tag{A1}$$

where $\{M_x^{(\theta)}\}_{x\in\{0,1\}^n}$ is the input-picture POVM. Define

$$\mathrm{TVD}(p,q) = \frac{1}{2}\sum_x |p(x) - q(x)|, \tag{A2}$$

$$D_{\mathrm{tr}}(\rho,\sigma) = \frac{1}{2}\|\rho - \sigma\|_1. \tag{A3}$$

For finite probability spaces,

$$\mathrm{TVD}(p,q) = \max_A |p(A) - q(A)|. \tag{A4}$$

Lemma 1 (Measurement contracts distinguishability)**.** *Every quantum-to-classical channel $\mathcal{M}$ satisfies*

$$\mathrm{TVD}(\mathcal{M}(\rho), \mathcal{M}(\sigma)) \le D_{\mathrm{tr}}(\rho,\sigma). \tag{A5}$$

Proof. For an output event A, set $M_A = \sum_{x\in A} M_x$. The POVM conditions give $0 \le M_A \le I$. The variational characterization of trace distance is

$$D_{\mathrm{tr}}(\rho,\sigma) = \max_{0\le F\le I} |\mathrm{Tr}[F(\rho-\sigma)]| \tag{A6}$$

[15]. Therefore,

$$\mathrm{TVD}(\mathcal{M}(\rho), \mathcal{M}(\sigma)) = \max_A |\mathrm{Tr}[M_A(\rho-\sigma)]| \tag{A7}$$

$$\le \max_{0\le F\le I} |\mathrm{Tr}[F(\rho-\sigma)]|, \tag{A8}$$

which proves Eq. (A5). □

Theorem 1 (Pairwise approximation lower bound)**.** *For any target conditional law p, model channel $\mathcal{M}_\theta$, and conditions c, c',*

$$\max\left\{\mathrm{TVD}(q_\theta(\cdot \mid c), p(\cdot \mid c)), \mathrm{TVD}(q_\theta(\cdot \mid c'), p(\cdot \mid c'))\right\} \ge \frac{1}{2}\left[\mathrm{TVD}(p(\cdot \mid c), p(\cdot \mid c')) - D_{\mathrm{tr}}(\rho_c, \rho_{c'})\right]_+. \tag{A9}$$

Proof. Let $e_c = \mathrm{TVD}(q_\theta(\cdot \mid c), p(\cdot \mid c))$, and define $e_{c'}$ similarly. Two applications of the triangle inequality give

$$\mathrm{TVD}(p(\cdot \mid c), p(\cdot \mid c')) \le e_c + \mathrm{TVD}(q_\theta(\cdot \mid c), q_\theta(\cdot \mid c')) + e_{c'} \tag{A10}$$

$$\le e_c + D_{\mathrm{tr}}(\rho_c, \rho_{c'}) + e_{c'} \tag{A11}$$

$$\le 2\max\{e_c, e_{c'}\} + D_{\mathrm{tr}}(\rho_c, \rho_{c'}), \tag{A12}$$

where the second line uses Lemma 1. Rearrangement and nonnegativity prove Eq. (A9). □

2. Exact and resource-scaled no-go results

Corollary 1 (Exact universality requires orthogonal condition states)**.** *If a fixed input encoding and a trainable channel family represent every conditional law, then*

$$D_{\mathrm{tr}}(\rho_c, \rho_{c'}) = 1 \quad \textit{for every } c \neq c'. \tag{A13}$$

Equivalently, every pair of input states has orthogonal support.

Proof. For any selected pair $c \neq c'$, choose a target with $p(\cdot \mid c) = \delta_{0^n}$ and $p(\cdot \mid c') = \delta_{1^n}$. The endpoint laws have total-variation distance one. Exact representation makes the left side of Eq. (A9) zero, so $D_{\mathrm{tr}}(\rho_c, \rho_{c'}) \geq 1$. Trace distance between states is at most one. Equality holds exactly when their supports are orthogonal. □

The simplest classically tractable counterexample is a constant product encoding, $\rho_c = |0^N\rangle\langle 0^N|$ for every c. Equation (A1) then gives a condition-independent output for every circuit and every parameter choice. Efficient basis sampling and amplitude evaluation alone therefore imply no conditional expressivity.

For an asymptotic claim, index the encoder and circuit resources by R and write the input as $\rho_c^{(R)}$. If the encoder also has trainable shared parameters, include them in R.

Theorem 2 (Universality forces asymptotic label separation)**.** *Suppose a resource-scaled family uniformly approximates every total-variation-continuous conditional law on $\mathcal{C} = [0, 1]$. Then*

$$\sup_R D_{\mathrm{tr}}\left(\rho_c^{(R)}, \rho_{c'}^{(R)}\right) = 1 \quad \textit{for every } c \neq c'. \tag{A14}$$

Approximating a suitable continuous target with uniform error ε requires

$$D_{\mathrm{tr}}\left(\rho_c^{(R)}, \rho_{c'}^{(R)}\right) \geq 1 - 2\varepsilon. \tag{A15}$$

Proof. Assume $c < c'$. Define a continuous function $\lambda : [0, 1] \to [0, 1]$ that is zero on $[0, c]$, linear from zero to one on $[c, c']$, and one on $[c', 1]$. The conditional law

$$p(\cdot \mid t) = (1 - \lambda(t))\delta_{0^n} + \lambda(t)\delta_{1^n} \tag{A16}$$

is continuous in total variation, and its laws at c, c' have distance one. Apply Theorem 1 to a resource that approximates this target with uniform error ε. This gives Eq. (A15). Letting ε tend to zero proves Eq. (A14). □

Equation (A14) is the operational boundary between shared overlapping encodings and condition labels. Pairwise orthogonal supports, when attained, admit a zero-error measurement of the condition; when they are approached asymptotically, the optimal discrimination error tends to zero. Either case realizes condition routing rather than universality from a shared overlapping representation.

The obstruction is stronger when the encoder family has a resource-independent continuity modulus. Suppose

$$D_{\mathrm{tr}}\left(\rho_c^{(R)}, \rho_{c'}^{(R)}\right) \leq \omega(|c - c'|) \quad \text{for every } R, c, c', \tag{A17}$$

where $\omega(h) \to 0$ as $h \to 0$. Choose h with $\omega(h) \leq 1/2$ and place the ramp in Eq. (A16) across an interval of width h. Theorem 1 gives the resource-independent bound

$$\sup_c \mathrm{TVD}(q_\theta(\cdot \mid c), p(\cdot \mid c)) \geq \frac{1}{2}(1 - \omega(h)) \geq \frac{1}{4}. \tag{A18}$$

Thus a universal family must lose every continuity modulus that is uniform over resource size.

Theorem 1 also gives a necessary condition for any restricted positive result. Uniform error at most ε implies

$$\mathrm{TVD}(p(\cdot \mid c), p(\cdot \mid c')) \leq D_{\mathrm{tr}}(\rho_c, \rho_{c'}) + 2\varepsilon \quad \text{for all } c, c'. \tag{A19}$$

This condition is necessary, not sufficient. It does not account for locality, parameter count, or Fourier bandwidth.

3. Discarded selectors are branch mixtures

Let S be a selector register and let R contain the other model registers. Immediately before the IQP diagonal, suppose the joint state factorizes as $\sigma_S(c)\otimes\tau_R$, with only $\sigma_S(c)$ depending on the condition. Every computational-basis diagonal has the controlled form

$$D = \sum_{s\in\{0,1\}^r} |s\rangle\langle s|_S \otimes D_s, \tag{A20}$$

where every D_s is diagonal on R.

Proposition 2 (Discarded-selector mixture identity)**.** *Applying D and tracing out S gives*

$$\mathrm{Tr}_S\left[D(\sigma_S(c)\otimes\tau_R)D^\dagger\right] = \sum_s w_c(s) D_s \tau_R D_s^\dagger, \qquad w_c(s) = \langle s|\sigma_S(c)|s\rangle. \tag{A21}$$

After the remaining IQP operations and visible measurement,

$$q(x \mid c) = \sum_s w_c(s) q_s(x), \tag{A22}$$

where q_s is independent of c.

Proof. Expand $\sigma_S(c) = \sum_{s,t} \sigma_{st}(c)\,|s\rangle\langle t|$. Then

$$D(\sigma_S(c)\otimes\tau_R)D^\dagger = \sum_{s,t} \sigma_{st}(c)\,|s\rangle\langle t| \otimes D_s\tau_R D_t^\dagger. \tag{A23}$$

The identity $\mathrm{Tr}_S(|s\rangle\langle t|) = \delta_{st}$ removes every off-diagonal term and proves Eq. (A21). Linearity of the remaining channel proves Eq. (A22). A final Hadamard on S does not alter its partial trace. □

The proposition covers one-hot selectors, binary address registers, and independent Bernoulli selectors whose weights form Bernstein polynomials. They may be compiled into one large diagonal, but the visible law remains a mixture of condition-independent branches. The factorization assumption is needed for this identity. An input that entangles S with R need not obey Eq. (A22), but it still obeys Theorem 1.

4. Broadcast product encoder

Consider N product rotations,

$$|\Phi(c)\rangle = \bigotimes_{j=1}^{N} R_Y(\alpha_j(c))\,|0\rangle\,, \qquad R_Y(a)\,|0\rangle = \cos(a/2)\,|0\rangle + \sin(a/2)\,|1\rangle\,. \tag{A24}$$

For pure states,

$$D_{\mathrm{tr}}(|\psi\rangle\,,|\varphi\rangle) = \sqrt{1-|\langle\psi|\varphi\rangle|^2}, \tag{A25}$$

where ket arguments abbreviate the corresponding projectors. This follows by diagonalizing $|\psi\rangle\langle\psi| - |\varphi\rangle\langle\varphi|$, whose nonzero eigenvalues are $\pm\sqrt{1-|\langle\psi|\varphi\rangle|^2}$. Since

$$\langle 0|R_Y(a)^\dagger R_Y(b)|0\rangle = \cos\left(\frac{a-b}{2}\right), \tag{A26}$$

the exact product-state distance is

$$d_N(c,c') = \sqrt{1-\prod_{j=1}^{N}\cos^2\left(\frac{\alpha_j(c)-\alpha_j(c')}{2}\right)}. \tag{A27}$$

For the broadcast map $\alpha_j(c) = \alpha(c)$,

$$d_N(c, c') = \sqrt{1 - \cos^{2N}\left(\frac{\alpha(c) - \alpha(c')}{2}\right)}. \tag{A28}$$

At fixed N, any pair with $|\alpha(c) - \alpha(c')| < \pi$ remains nonorthogonal, so exact universality is impossible.

Suppose α is L_α-Lipschitz. Put $u = \cos^2(\Delta\alpha/2)$. The inequalities $1 - u^N \le N(1-u)$ and $|\sin x| \le |x|$ give

$$\begin{aligned} d_N(c, c') &= \sqrt{1 - u^N} && \text{(A29)}\\ &\le \sqrt{N(1-u)} && \text{(A30)}\\ &= \sqrt{N}\left|\sin\left(\frac{\Delta\alpha}{2}\right)\right| && \text{(A31)}\\ &\le \frac{\sqrt{N}L_\alpha}{2}|c - c'|. && \text{(A32)} \end{aligned}$$

For the continuous ramp target over an interval of width h, Theorem 1 therefore yields

$$\sup_c \mathrm{TVD}(q_\theta(\cdot \mid c), p(\cdot \mid c)) \ge \frac{1}{2}\left(1 - \frac{\sqrt{N}L_\alpha h}{2}\right)_+. \tag{A33}$$

The Gaussian encoder in Eq. (5) has $N = 19$ and $L_\alpha = 0.70\pi$. With $h = 0.01$,

$$\frac{1}{2}\left(1 - \frac{\sqrt{19}(0.70\pi)(0.01)}{2}\right) = 0.4760357\ldots. \tag{A34}$$

This bound holds for every downstream quantum circuit, not only an IQP diagonal.

For any fixed $c \neq c'$ with $0 < |\alpha(c) - \alpha(c')| < \pi$, Eq. (A28) tends to one as $N \to \infty$. Repeated encoding evades the fixed-pair obstruction only through asymptotic orthogonalization. Equation (A32) shows the same effect locally: the common trace-distance modulus grows as $\sqrt{N}$.

5. Fixed Fourier and parameter-dimension obstructions

The one-qubit broadcast state has density operator

$$R_Y(\alpha)\,|0\rangle\langle 0|\,R_Y(\alpha)^\dagger = \frac{1}{2}\left(I + \sin(\alpha)X + \cos(\alpha)Z\right). \tag{A35}$$

Every entry of its N-fold tensor power is a trigonometric polynomial of degree at most N. For any fixed downstream channel and output event A,

$$q_\theta(A \mid \alpha) = \mathrm{Tr}[M_A\,|\Phi(\alpha)\rangle\langle\Phi(\alpha)|] \tag{A36}$$

therefore has degree at most N [11, 12].

Proposition 3 (Fourier counterexample)**.** *On a full 2π period, define*

$$p(0^n \mid \alpha) = \frac{1 + \eta\cos((N+1)\alpha)}{2}, \qquad p(1^n \mid \alpha) = 1 - p(0^n \mid \alpha), \tag{A37}$$

where $0 < \eta \le 1$. Every fixed channel fed by the N-copy broadcast encoder satisfies

$$\sup_\alpha \mathrm{TVD}(q_\theta(\cdot \mid \alpha), p(\cdot \mid \alpha)) \ge \frac{\eta}{4}. \tag{A38}$$

Proof. Let $g(\alpha) = q_\theta(0^n \mid \alpha)$ and $f(\alpha) = p(0^n \mid \alpha)$. The $(N+1)$st Fourier coefficient of g is zero, while that of f is $\eta/4$. If $\|g - f\|_\infty < \eta/4$, then every Fourier coefficient of $g - f$ has magnitude below $\eta/4$, a contradiction. Thus $\|g - f\|_\infty \ge \eta/4$. Total variation dominates the probability difference of the event $\{0^n\}$. □

The full-period domain is needed only for the explicit constant. On every closed interval I of nonzero length, degree-N trigonometric polynomials form a finite-dimensional closed proper subspace $\mathcal{T}_N$ of $C(I)$. The target in Eq. (A37) does not belong to this subspace because equality on an interval would extend analytically to the real line and contradict its nonzero $(N+1)$st Fourier coefficient. Hence

$$\inf_{g\in\mathcal{T}_N}\left\|g-\frac{1+\eta\cos((N+1)\,\cdot)}{2}\right\|_{\infty,I}>0. \tag{A39}$$

Fixed bandwidth is therefore not dense on the Gaussian angle interval.

For a separate obstruction, fix K encoded conditions and an n-bit visible output. Removing one redundant coordinate from each probability simplex gives target-table dimension

$$d=K(2^n-1). \tag{A40}$$

An IQP model with M periodic phases defines a real-analytic map $F:\mathbb{T}^M\to\mathbb{R}^d$.

Proposition 4 (Parameter-dimension obstruction)**.** *If $M<d$, the image of F has measure zero and is not dense in the product of the K probability simplices.*

Proof. The derivative of F has rank at most $M<d$ everywhere. Sard's theorem therefore gives a measure-zero image. Since $\mathbb{T}^M$ is compact and F is continuous, the image is compact and closed. A closed measure-zero subset cannot be dense in a simplex product with nonempty relative interior. □

A complete two-local diagonal on Q qubits has $M=Q+\binom{Q}{2}$. The Gaussian setting has $M=190<5(2^{19}-1)=2{,}621{,}435$. The 13-qubit Curie–Weiss setting has $M=91<6(2^8-1)=1{,}530$ when evaluated on its six training temperatures. This count rules out density in arbitrary conditional tables at those inputs. It does not claim that the experimental target families occupy all table dimensions.

6. Interpretation

Classical tractability is an algorithmic property of the input description. It controls basis sampling and amplitude-ratio evaluation, but it cannot increase $D_{\mathrm{tr}}(\rho_c,\rho_{c'})$. Constant product states and orthogonal computational-basis labels are both classically tractable, yet the first removes all condition dependence and the second enables exact routing.

The no-go theorem does not rule out a positive approximation theorem for a restricted target class. Such a theorem must at least satisfy Eq. (A19), and it must also respect the diagonal locality, Fourier bandwidth, and parameter count. Whether the complete two-local IQP family is universal for a useful polynomial-resource subclass of encoding-compatible conditional laws remains open.

Appendix B: Analytical details

1. Finite conditional response

Let the product-state angle map be affine, $\alpha(c)=b+Wc$, where row i of W is w_i^{T}. Expanding each local sine and cosine into exponentials shows that every amplitude has frequencies in signed sums of the rows of W. Squaring an amplitude gives

$$\Omega_{\mathrm{full}}(W)\subseteq\left\{\sum_{i=1}^{n}s_iw_i:s_i\in\{-1,0,1\}\right\}. \tag{B1}$$

Coincident frequencies and vanishing coefficients can make the inclusion strict. The argument establishes a finite response space at fixed encoding width; it does not provide an efficient coefficient-evaluation procedure or a universal-approximation result.

For this expansion, enumerate the vertex and edge terms of G by binary masks $g_j\in\{0,1\}^n$, with $Z_{g_j}=\bigotimes_i Z_i^{g_{j,i}}$. For an observable Z_a, the active-generator expansion retains only terms with $a\cdot g_j=1 \pmod 2$. Grouping active subsets by their XOR mask h and using $\langle Y_i\rangle=0$ for a real R_Y product input removes terms with $h\wedge a\neq 0$. Observable responses can therefore have strictly smaller frequency support than full probabilities. This is a direct specialization of established circuit-Fourier and IQP active-generator descriptions [8, 11, 12].

2. Product and block-product estimator

This derivation specializes Van den Nest's CT–ECS matrix-element estimator to the one-sparse operator obtained by conjugating a Pauli character through an IQP diagonal [6, Definition 1 and Theorems 1 and 3]. Write the tractable reference amplitude as $\psi_c(z) = \langle z|H^{\otimes n}|\phi(c)\rangle$. The classical observable estimator samples z from $|\psi_c(z)|^2$ and evaluates an amplitude ratio $R_c(z,a) = \psi_c(z\oplus a)/\psi_c(z)$ together with the phase factor generated by the active diagonal terms. The ratio obeys

$$\mathbb{E}_z|R_c(z,a)|^2 = \sum_{z:|\psi_c(z)|>0} |\psi_c(z)|^2 \frac{|\psi_c(z\oplus a)|^2}{|\psi_c(z)|^2} \tag{B2}$$

$$\leq \sum_z |\psi_c(z\oplus a)|^2 = 1. \tag{B3}$$

The equality used in the main text holds when the shifted support is fully covered; the inequality is the general statement when some reference amplitudes vanish. In either case the variance is at most one. The bound is on the estimator variance, not on optimizer convergence or raw gradient magnitude.

The same argument applies to a block-product input $|\phi(c)\rangle = \bigotimes_B |\phi_B(c)\rangle$. The ratio factorizes over blocks, and normalization gives

$$\mathbb{E}|R_c(z,a)|^2 \leq \prod_B \sum_{z_B} |\phi_B(z_B\oplus a_B)|^2 = 1. \tag{B4}$$

This observation supports the archived block-encoding study in Sec. F. The Gaussian production experiments use single-qubit product states, while the Curie–Weiss production experiments use the spin-flip two-branch state, whose radial amplitude also factorizes over Hamming shells and is classically tractable.

3. Hamming-kernel character loss

For $a_i \sim \text{Bernoulli}(q_\sigma)$ with q_σ defined in Eq. (20),

$$\mathbb{E}_a[(-1)^{a\cdot(x\oplus y)}] = \prod_i \left[(1-q_\sigma) + q_\sigma(-1)^{x_i\oplus y_i}\right] \tag{B5}$$

$$= \exp[-d_H(x,y)/(2\sigma^2)]. \tag{B6}$$

For condition set C, the training objective is the balanced average

$$\mathcal{L}(\theta) = \frac{1}{|C|}\sum_{c\in C} \text{MMD}^2(q_\theta(\cdot\mid c), p(\cdot\mid c)). \tag{B7}$$

Independent model and target draws give the standard unbiased U-statistic. All random streams for masks, model samples, target batches, and validation are derived separately and recorded in the campaign artifacts.

Appendix C: Gaussian protocol and complete factorial summary

For each n, width factor s, and condition c, the target shell law is a truncated discrete Gaussian. Its untruncated location is calibrated so that the truncated mean is exactly $n(0.15+0.70c)$, and its nominal width is $s\sqrt{n}$. Conditional and no-condition models share the complete two-local interaction graph, initialization, datasets, optimizer seed, minibatches, and estimator streams. Single-condition references use condition-tagged streams.

The factorial contains $4\times 3\times 10$ seed cells and 840 atomic fits. The condition-count study adds 180 fits for a total of 1,020. Training uses 10,000 strings and validation uses 2,000 strings per condition in the fixed-per-condition regime. The fixed-total regime divides 18,000 training and 3,600 validation strings across conditions using prefixes of the same master pools. Held-out conditions have no sample role in the stored datasets.

Table I reports every size–width profile-TVD effect. Positive no-condition effects favor conditioning. Positive adjacent-reference or mixture effects would favor the shared model.

Table II shows the condition-density result used in the main text. The full per-condition rows, worst-condition values, paired intervals, and exact probability vectors remain in the machine-readable report.

Profile TVD intentionally quotients out within-shell variation. The exact raw-bitstring report shows that low profile error does not imply a uniform law inside each shell. We retain the raw probabilities and all secondary metrics in the release artifacts rather than reproducing hundreds of diagnostic rows in the PDF.

TABLE I: Gaussian factorial profile-TVD effects with paired 95% bootstrap intervals. Status letters are S for supported, M for mixed, and U for unsupported, in the order no-condition, adjacent-reference, mixture.

n	s	no condition − conditioned	adjacent reference − conditioned	mixture − conditioned	status
8	0.35	+0.2402 [+0.2278, +0.2536]	-0.0292 [-0.0468, -0.0108]	-0.0248 [-0.0418, -0.0064]	S/U/U
8	0.50	+0.2170 [+0.2015, +0.2418]	-0.0280 [-0.0336, -0.0225]	-0.0363 [-0.0432, -0.0292]	S/U/U
8	0.75	+0.1460 [+0.1293, +0.1714]	-0.0276 [-0.0339, -0.0214]	-0.0525 [-0.0584, -0.0461]	S/U/U
12	0.35	+0.2419 [+0.2352, +0.2505]	-0.0221 [-0.0255, -0.0190]	-0.0016 [-0.0052, +0.0019]	S/U/M
12	0.50	+0.2291 [+0.2213, +0.2391]	-0.0180 [-0.0213, -0.0146]	-0.0251 [-0.0283, -0.0216]	S/U/U
12	0.75	+0.1608 [+0.1515, +0.1722]	-0.0129 [-0.0151, -0.0107]	-0.0548 [-0.0567, -0.0526]	S/U/U
16	0.35	+0.2472 [+0.2421, +0.2519]	-0.0229 [-0.0279, -0.0174]	+0.0117 [+0.0067, +0.0172]	S/U/S
16	0.50	+0.2440 [+0.2354, +0.2537]	-0.0040 [-0.0071, +0.0001]	-0.0099 [-0.0133, -0.0057]	S/M/U
16	0.75	+0.1667 [+0.1570, +0.1770]	-0.0192 [-0.0205, -0.0178]	-0.0681 [-0.0696, -0.0666]	S/U/U
19	0.35	+0.2620 [+0.2594, +0.2645]	-0.0217 [-0.0240, -0.0192]	+0.0188 [+0.0164, +0.0213]	S/U/S
19	0.50	+0.2668 [+0.2587, +0.2737]	+0.0016 [-0.0012, +0.0043]	-0.0029 [-0.0053, -0.0006]	S/M/U
19	0.75	+0.1807 [+0.1648, +0.1999]	-0.0111 [-0.0165, -0.0049]	-0.0619 [-0.0670, -0.0560]	S/U/U

TABLE II: Condition-count study at $n = 19, s = 0.50$. Mean and worst are held-out profile TVD. The shared model keeps 190 phases and 2,000 updates.

K	regime	train/condition	validation/condition	mean	worst	phases	shared updates	reference updates
2	fixed per condition	10,000	2,000	0.3819	0.5170	190	2,000	4,000
3	fixed per condition	10,000	2,000	0.3402	0.4489	190	2,000	6,000
5	fixed per condition	10,000	2,000	0.1673	0.2917	190	2,000	10,000
9	fixed per condition	10,000	2,000	0.1595	0.2947	190	2,000	18,000
2	fixed total	9,000	1,800	0.3819	0.5166	190	2,000	4,000
3	fixed total	6,000	1,200	0.3399	0.4488	190	2,000	6,000
5	fixed total	3,600	720	0.1645	0.2932	190	2,000	10,000
9	fixed total	2,000	400	0.1622	0.2936	190	2,000	18,000

Appendix D: Curie–Weiss protocol and results

For $s_i = 2x_i - 1$, $M = \sum_i s_i$, and $n = 8$, the exact target is

$$p_\beta(x) = Z_\beta^{-1} \exp\left(\frac{\beta M^2}{2n}\right). \tag{D1}$$

The campaign checks normalization, permutation invariance, and spin-flip invariance before writing a dataset. Each of ten dataset archives contains 10,000 training and 2,000 validation strings at the six training temperatures. Exact laws at held-out midpoints enter only after checkpoint selection.

The v3 broadcast map is piecewise linear. Its knots were chosen from development screens before production and are recorded in the manifest. The map is not presented as target-independent. The production model uses the spin-flip two-branch state on eight visible and five ancilla inputs, applies a complete two-local 13-qubit IQP unitary, and sums the exact joint law over all ancilla outcomes. The encoding family and map were frozen before the 80 production fits.

Table III gives the complete aggregate comparison set. Temperatures are averaged within seed before bootstrapping. Positive effects favor the named conditioned model. The shared model matches the adjacent mixture and improves on the target-aware adjacent oracle in this campaign; both profile and raw TVD improve relative to no conditioning.

The five-ancilla choice followed a separate three-seed development study at $r \in \{0, 1, 3\}$. It motivated adding latent capacity but did not test $r = 5$, establish an optimal ancilla count, or isolate mixed-state rank from parameter count.

1. Latent-input construction and development protocol

The latent-input study is separate from the immutable v3 campaign. It uses the target-independent coordinate $t = (\beta - 0.5)/(1.5 - 0.5)$ and $r \in \{1, 3, 5\}$ ancillas. Each ancilla is prepared with computational-basis probability

TABLE III: All paired Curie–Weiss v3 effects.

comparison	effect [95% interval]	status
cold_no_condition_bitstring_tvd	+0.209 [+0.202, +0.215]	supported
cold_no_condition_profile_tvd	+0.212 [+0.205, +0.219]	supported
conditional_r5_bitstring_tvd_held_out	+0.128 [+0.123, +0.134]	supported
conditional_r5_bitstring_tvd_training	+0.140 [+0.135, +0.145]	supported
conditional_r5_profile_tvd_held_out	+0.121 [+0.116, +0.127]	supported
conditional_r5_profile_tvd_training	+0.134 [+0.129, +0.140]	supported
critical_no_condition_bitstring_tvd	+0.056 [+0.048, +0.065]	supported
critical_no_condition_profile_tvd	+0.038 [+0.030, +0.047]	supported
r5_vs_mixture_bitstring_tvd_held_out	+0.008 [+0.002, +0.013]	supported
r5_vs_mixture_profile_tvd_held_out	-0.011 [-0.017, -0.005]	unsupported
r5_vs_profile_oracle_held_out	+0.015 [+0.011, +0.019]	supported
r5_vs_raw_oracle_held_out	+0.036 [+0.031, +0.041]	supported
r5_vs_reference_bitstring_tvd_training	+0.008 [+0.003, +0.013]	supported
r5_vs_reference_profile_tvd_training	-0.020 [-0.024, -0.015]	unsupported

TABLE IV: Ancilla development study. Entries are mean held-out TVD.

r	model	profile TVD	raw TVD
0	conditioned	0.208	0.321
0	no condition	0.221	0.298
1	conditioned	0.166	0.213
1	no condition	0.227	0.261
3	conditioned	0.111	0.128
3	no condition	0.155	0.175

$p(z_k = 1 \mid t) = t$. Learned same-axis controlled rotations then produce branch angles

$$\alpha_i(t, z) = \pi(0.35 - 0.70t) + \sum_{k=1}^{r} L_{ik}(-1)^{z_k}. \tag{D2}$$

The diagonal contains only the 36 complete two-local visible generators and one shared phase vector θ. It never acts on an ancilla. Tracing the ancillas gives the exact visible marginal

$$q_{\theta,L}(x \mid t) = \sum_{z \in \{0,1\}^r} t^{|z|}(1-t)^{r-|z|} q_\theta^{(v)}[x \mid \alpha(t, z)]. \tag{D3}$$

Thus the marginal is a mixture over input states, as every traced-ancilla construction must be, but not a phase-multiplexed bank of IQP experts. Unit tests compare this marginal with direct branch evaluation and require agreement within 10^{-12}; setting $L = 0$ recovers the ordinary shared visible circuit to the same tolerance.

For a visible mask, joint-state sampling first draws z and then draws a visible product string conditioned on z. The amplitude ratio leaves z fixed, so the Bernoulli amplitude cancels and the remaining ratio factorizes over visible qubits. The Gaussian-MMD estimator therefore remains tractable without summing over 2^r branches. At $n = 8, r = 5$, the model has 36 shared IQP phases, 40 controlled-input couplings, and 13 qubits. Its diagonal still has 36 rotations and maximum locality two.

The critical-gap protocol trains at $\beta \in \{0.5, 0.7, 0.8, 1.2, 1.3, 1.5\}$ and evaluates $\{0.9, 1.0, 1.1\}$. The hot-to-cold protocol trains at $\{0.5, 0.6, 0.7, 0.8, 0.9\}$ and evaluates $\{1.0, 1.1, 1.2, 1.3, 1.4, 1.5\}$. Both use exact multiscale population MMD, 10,000 Adam updates, and ten optimizer seeds for the headline comparisons. Test targets are constructed after the minimum training-objective checkpoint is fixed. The leakage audit records that the encoding uses no target samples, Curie–Weiss energy, target symmetry, known critical point, or test outcome.

TABLE V: Exploratory ancilla-rank, matched-capacity, and non-Curie controls. Entries are mean held-out errors over optimizer seeds. The final column is the absolute error in normalized magnetization magnitude; lower is better.

control	parameters	profile TVD	raw TVD	$\mathbb{E}\lvert m\rvert$ error
Curie–Weiss affine	36	0.262	0.341	0.055
Latent input, $r = 1$	44	0.108	0.225	0.050
Latent input, $r = 3$	60	0.074	0.161	0.033
Latent input, $r = 5$	76	0.060	0.116	0.027
Visible-only capacity	76	0.144	0.294	0.031
Gaussian affine	36	0.110	0.241	0.016
Gaussian latent input, $r = 5$	76	0.071	0.103	0.005
Gaussian visible-only capacity	76	0.126	0.320	0.003

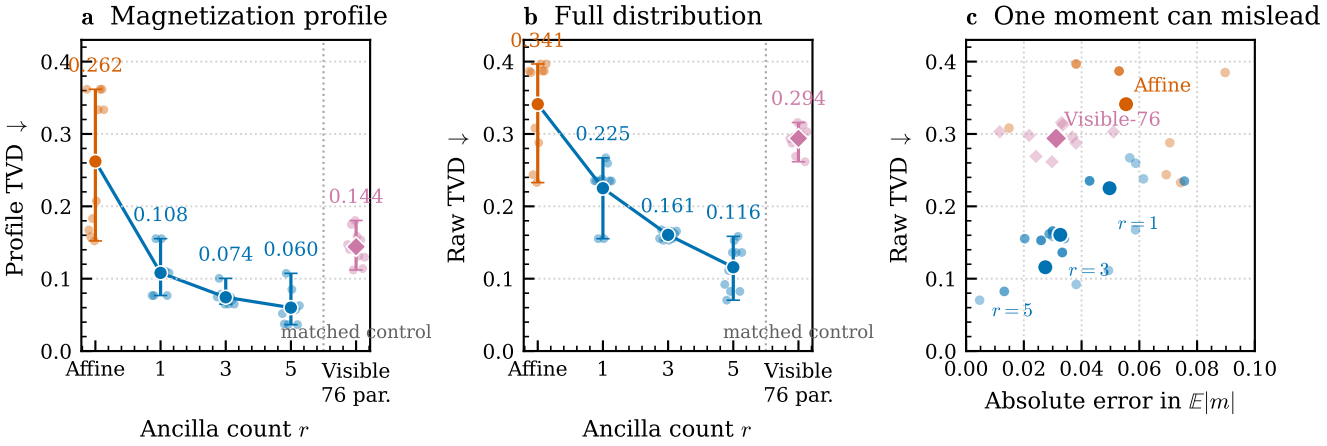


FIG. 2: Seed-level distributional audit of the target-independent critical-gap prototype. Each faint point is one optimizer seed averaged over the three held-out temperatures; large markers show the ten-seed mean and bars span the full seed range. Panels **a** and **b** report magnetization-profile and raw TVD. The line connects the affine baseline to the $r = 1, 3, 5$ ancilla ladder; the diamond is the 76-parameter visible-only control matched to $r = 5$. Panel **c** compares raw TVD with absolute error in $\mathbb{E}|m|$. The visible-only control matches this single moment about as well as $r = 5$ but remains inaccurate at the distribution level. All metrics use exact probabilities; MMD is the training objective, not the evaluation metric shown here. Lower is better on every axis.

Table V reports the ancilla ladder and the matched 76-parameter visible-only control. The Curie–Weiss entries use ten seeds and 10,000 updates. The non-Curie Gaussian location-path control uses five seeds and 5,000 updates and withholds the same normalized middle interval. Its target mean follows the existing generic path $\mathbb{E}[K \mid t]/n = 0.15 + 0.70t$ at fixed width $\sqrt{n}/2$.

The experiment is a capacity and optimization prototype rather than a new production campaign. Exact targets replace finite training samples, the encoder and ancilla ladder were chosen during exploration, and only eight visible spins are tested. The withheld critical interval measures generalization to absent conditions, but it remains within-range interpolation because training includes examples on both sides. The one-sided Curie–Weiss result is the extrapolation stress test: no target-independent method can identify an arbitrary unobserved phase from hot-phase data alone.

2. Historical v2 matched-capacity diagnostic

This diagnostic predates the v3 spin-flip campaign. It compares the v2 product-input ancilla model with a visible-only model containing every nonempty one-, two-, and three-qubit diagonal generator on eight visible qubits:

$$\binom{8}{1} + \binom{8}{2} + \binom{8}{3} = 8 + 28 + 56 = 92. \tag{D4}$$

TABLE VI: Matched-capacity paired effects. Positive values mean lower TVD for the five-ancilla model.

region	profile TVD	raw TVD
held out	+0.067 [+0.035, +0.095]	+0.144 [+0.105, +0.185]
critical	+0.118 [+0.082, +0.146]	+0.184 [+0.145, +0.218]
cold	+0.007 [-0.032, +0.048]	+0.082 [+0.033, +0.127]

This is one phase more than the five-ancilla complete two-local model. The control reuses the v2 datasets, product-input angle map, support-hashed initialization, estimator streams, 2,000-step optimizer, validation checkpoint rule, and exact evaluation grid. It does not reuse the v3 input state and is therefore neither a capacity match for v3 nor evidence that ancillas cause the v3 improvement. Critical aggregates average $\beta \in \{0.9, 1.0, 1.1\}$; cold aggregates average $\beta \in \{1.3, 1.4, 1.5\}$.

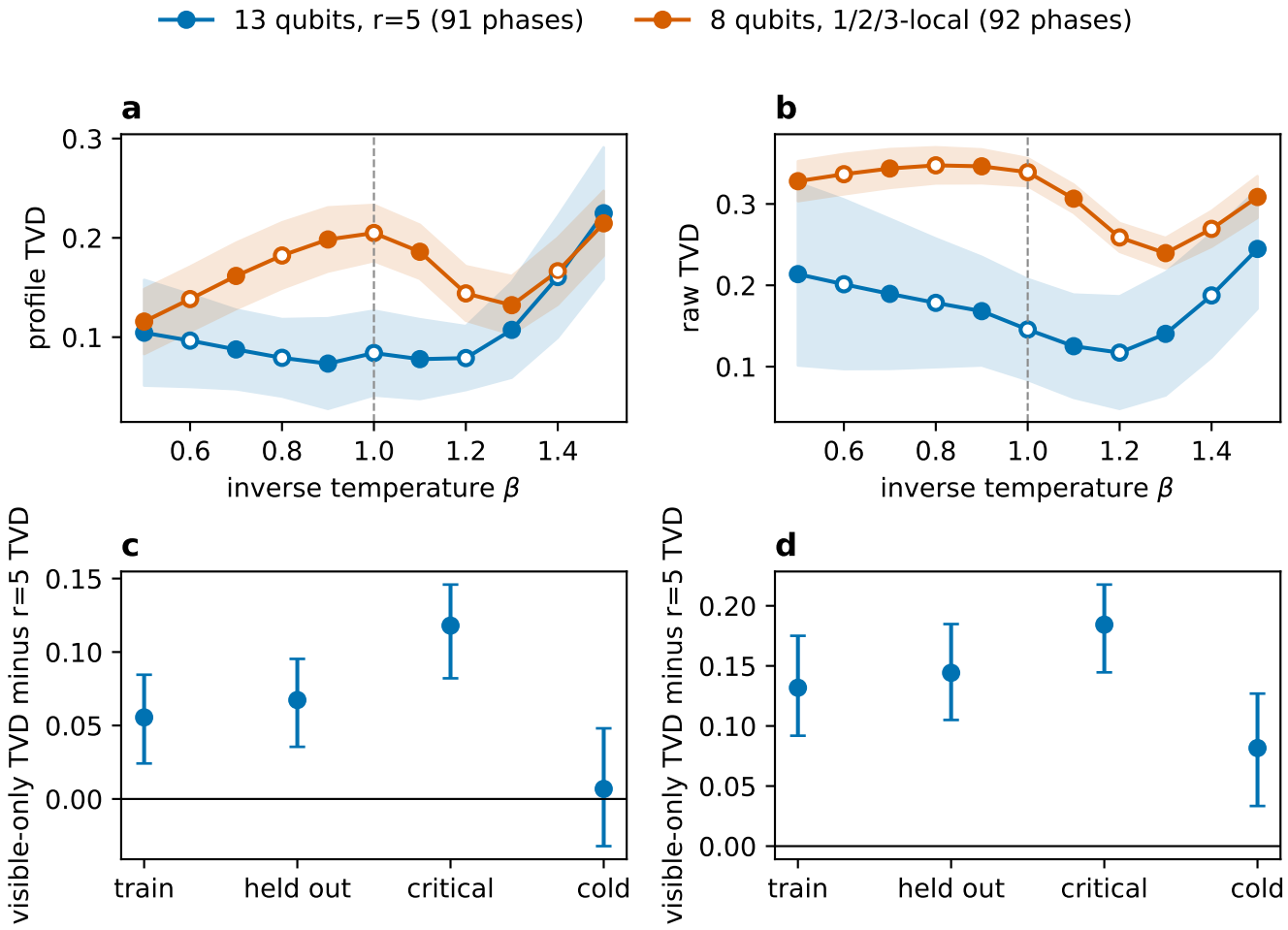


FIG. 3: Historical v2 matched-capacity diagnostic. Mean profile and raw TVD and paired regional effects compare the 91-phase v2 product-input ancilla model with the 92-phase visible-only model. Positive effects favor the v2 ancilla model. This is not a matched control for the v3 spin-flip model.

Appendix E: Exploratory symmetry screens

These post-core screens test whether matching an input state's symmetry to a target can improve finite-system fits. They were designed after the core campaigns and are not confirmatory evidence for the v3 result.

a. Two-block Z_2^2. A two-block Curie–Weiss target $p(s) \propto \exp[\beta(M_A^2 + M_B^2)/16]$ on two eight-spin blocks is invariant under independently flipping each block. The tested input is a product of two two-branch states, $|\phi\rangle = (|p_A\rangle^{\otimes 8} + X^{\otimes 8}|p_A\rangle^{\otimes 8}) \otimes (|p_B\rangle^{\otimes 8} + X^{\otimes 8}|p_B\rangle^{\otimes 8})$. Across ten matched seeds it outperforms the product and single-Z_2 inputs on the reported held-out metrics. This ordering is a screen result, not a general theorem about symmetry-matched encodings.

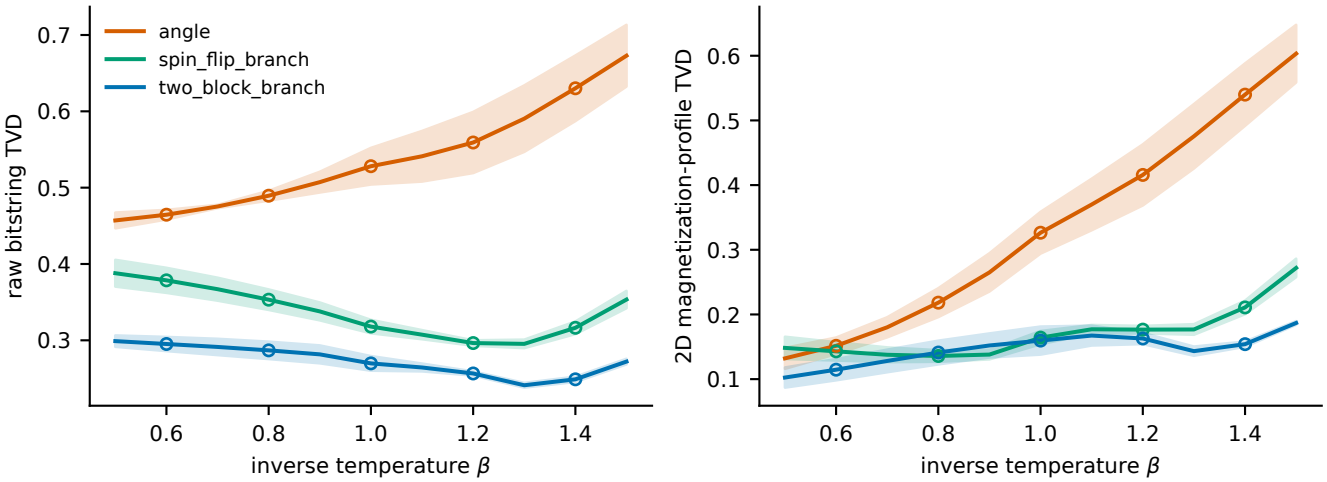


FIG. 4: Exploratory two-block Z_2^2 screen. Held-out raw and two-block magnetization-profile TVD for product, single-Z_2, and two-block-Z_2^2 inputs. Bands are one standard error across ten matched seeds; hollow markers denote held-out temperatures.

b. Asymmetric exchangeable targets. A radial-shell state with a frozen quartic shell law accepts two coordinates (c_1, c_2) and preserves permutation symmetry without enforcing spin flip. It improves the weight-profile and permutation-averaged raw TVD in the reported screen, while its unprocessed raw TVD is worse. The product and two-branch baselines receive only c_1, whereas the radial model receives both c_1 and c_2; the result therefore does not isolate symmetry matching from access to an additional condition coordinate.

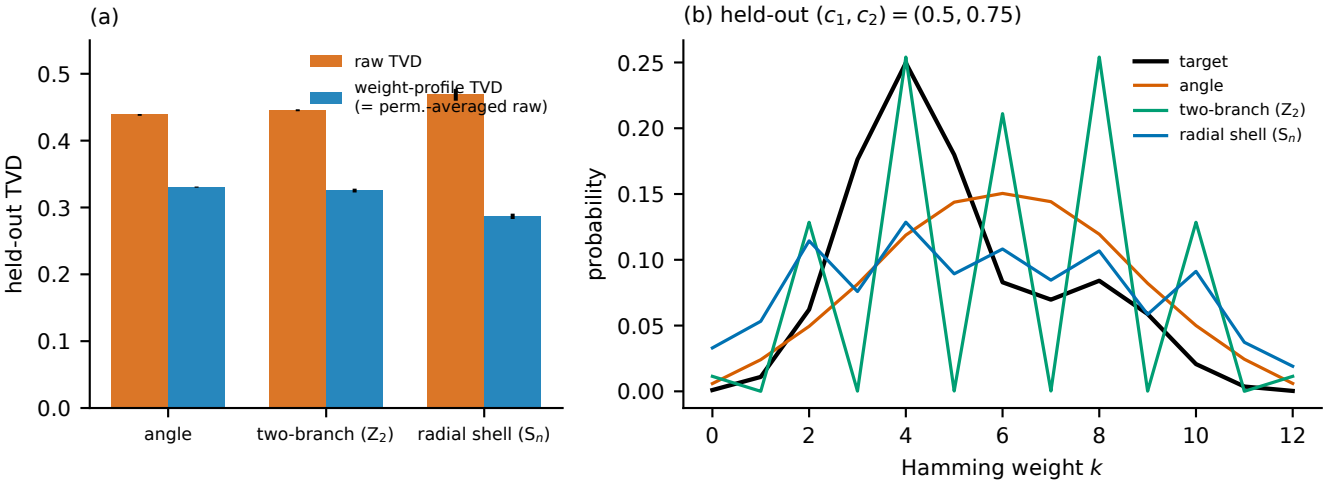


FIG. 5: Exploratory asymmetric radial-shell screen. Held-out raw and weight-profile TVD and representative weight profiles are shown for the three input families. The unequal condition information makes this a feasibility screen rather than a controlled encoder comparison.

Appendix F: Archived encoding comparison

We tested two extensions of the single-angle product encoding on the $n = 19, s = 0.50$ Gaussian task: noncommuting single-qubit reuploading and two-qubit block-product inputs. None improved the tuned production encoder. The study is supplementary because it followed a different development protocol and its alternatives were not used in either production campaign.

TABLE VII: Archived matched encoding comparison on the $n = 19, s = 0.50$ Gaussian task. Lower is better. The paired profile effect is relative to the single-angle baseline; positive values are worse.

encoding	structure	held-out profile TVD	held-out raw TVD	paired profile effect
single angle	$L = 1$	0.062 ± 0.004	0.348 ± 0.006	baseline
noncommuting	$L = 2$	0.072 ± 0.009	0.374 ± 0.027	$+0.011$ $[+0.004, +0.017]$
block ZZ	$\kappa = \pi/8$	0.070 ± 0.008	0.371 ± 0.020	$+0.009$ $[+0.003, +0.015]$
block ZZ	$\kappa = \pi/4$	0.071 ± 0.014	0.398 ± 0.009	$+0.009$ $[-0.001, +0.017]$

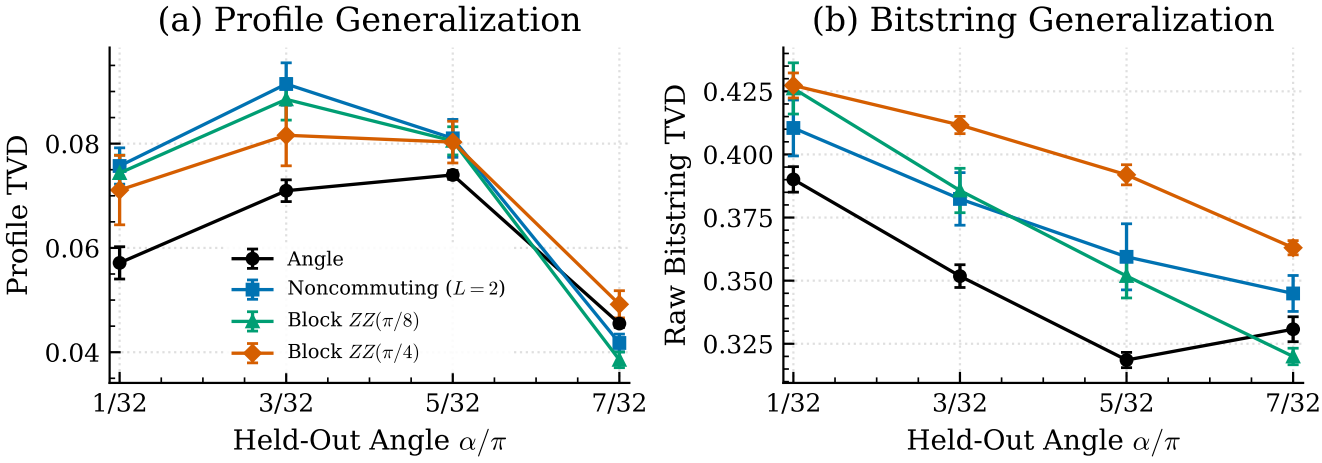


FIG. 6: Archived input-encoding comparison. The left panel reports held-out conditional MMD and the right panel reports raw profile TVD for the four tested architectures. Lower is better.

Appendix G: Finite estimator audit

The historical estimator audit contains 21 deterministic synthetic cases over qubit count, generator count, condition count, and condition dimension. It uses bounded-degree and complete two-local generator families at $n = 4$ and $n = 8$. Exact references are used where feasible; larger cases compare repeated estimator draws. The audit records expectation and gradient-estimator errors, runtime, memory, seeds, and content digests.

The confirmatory audit adds product, purified, and field-equivariant input states and uses sample budgets $S \in \{256, 1024, 4096, 16384\}$. For each input family, we compare estimated Pauli moments, MMD values, and phase gradients with exact enumeration on small systems. We report root-mean-square error as a function of S and compare its empirical slope with the Monte Carlo rate $S^{-1/2}$. Runtime and peak memory are recorded separately for sparse and complete interaction graphs. These comparisons validate the estimator interfaces used in the confirmatory campaign; they are not used to choose a reported training result.

This is finite consistency and resource evidence only. It does not support an asymptotic scaling claim, a statement about raw gradient magnitude, convergence of the optimizer, or absence of barren plateaus. No full training run is charged to this audit.

Appendix H: Negative and boundary studies

An eight-qubit pilot attempted joint conditioning on Gaussian location and width. Its frozen design required adequate single-condition representability before interpreting shared-model performance. Two narrow-width cells failed that screen, so the study stopped. It supplies no evidence for two-coordinate conditional generation and is retained as a negative result.

Older topology and ancilla studies remain archived under their original protocols. They are not pooled with either production campaign. The confirmatory topology ablation fixes $n = 19$, $w = 0.50$, and five training conditions, then compares complete, 4×5 rectangular-grid, and distance-three heavy-hex graphs under the common optimizer and estimator protocol. This is a robustness comparison between native graph families, not a parameter-matched test. The two symmetry screens above are supplementary exploratory evidence. Permutation twirling is included only as target-symmetry-aware post-processing; it is not a symmetry learned by the circuit.

Appendix I: Campaign integrity and reproduction

The Gaussian and Curie–Weiss stores bind every run identity to a manifest, role-tagged dataset, estimator streams, source digest, environment, code revision, checkpoint, exact probability archive, invocation, runtime, and peak memory. Stores reject undeclared identities, duplicate completion, and silent reuse of partial results. Report, claim, table, and figure generators require complete campaign state.

The Gaussian dependency closure includes a content-addressed snapshot of its historical execution source; the current live encoder is not substituted for that archived implementation.

The exploratory latent-input artifacts are stored separately from the immutable campaign stores. The SHA-256 digests for the $r = 1, 3, 5$ critical-gap files are `fd27fe18c156cd0bf1a7329a74f1a8c20f9b74f217d2909bd85ba0cce862b5cc`, `b4abc457cf4afa2b53276d3e5eb4c60cba686181ebed3efe65b260ca742903e9`, and `d8b5b9d725145865b9924ebca7cd64fae968400c8ca602352b664cc4247908eb`. The hot-to-cold, matched-visible, and Gaussian-control digests are `b5ab3cb31e2cce23132c9b1c85d80df0149f97d199cd7dceacb2ba61d157f9e0`, `6048a418e3ff5df806fa6e2103a8be76c333694a19f7233518ba6ab5e90fe15a`, and `888daf2f16758ee3281020d7fede681796a2d0fd4852c915eef91a24934b9a07`. Generated manuscript values are read from these JSON files rather than transcribed by hand.

The strict shared-phase implementation is isolated in `src/iqp_cond/latent_input.py`. Its tests verify normalized branch weights, recovery of the ordinary visible circuit when $L = 0$, and equality between the direct branch mixture and the ancilla-purified marginal within 10^{-12}. The prototype runner imports this implementation rather than maintaining a separate manuscript-specific evaluator.

Exhaustive per-seed and per-condition tables remain in JSON, NPZ, and generated TeX artifacts in the local review package. Keeping them machine-readable preserves the audit trail without turning the supplement into a second report.

After synchronizing the frozen environment, the full test suite, linter, phase-crossing interpolation generator, and `latexmk` reproduce the reported manuscript; the exact commands are recorded in the repository result README.

The confirmatory campaigns use one content-addressed source archive and one lockfile digest on both execution hosts. Run identities, rather than loop positions, determine the shards. Each host first benchmarks concurrency levels one, two, and four; the selected level must remain below 90 GiB peak aggregate memory and incur less than 20% per-job slowdown relative to serial execution. Result synchronization verifies file checksums, rejects duplicate run identities with different content, and permits summary generation only after every identity in the corresponding manifest is complete.